Standard Errors and Confidence Intervals for Total, Direct, and Indirect Effects in Continuous-Time Mediation Models

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Abstract

Mediation modeling using intensive longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive (DT-VAR) approaches are commonly used to estimate indirect effects in intensive longitudinal data (ILD), they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. To address these issues, continuous-time vector autoregressive (CT-VAR) models have been proposed as an alternative. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the total, direct, and indirect effects, for a range of time-intervals values, can be calculated using parameters estimated from CT-VAR models for causal inferential purposes. However, methods for calculating the uncertainty around the total, direct, and indirect effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results from using the delta and Monte Carlo methods to calculate standard errors and confidence intervals for the total, direct, and indirect effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed` `R` package.

Keywords: mediation, continuous-time, delta method, Monte Carlo method, R package, cTMed

Mediation is a widely adopted model across the social, behavioral, and health sciences. A recent Google Scholar search, conducted as of 24 January 2024, using search terms `"indirect effect OR mediation AND social science OR behavioral science OR health science"`, yielded over five million entries. While many mediation models still rely on cross-sectional data, there is a growing trend toward employing longitudinal designs, specifically cross-lagged panel designs. These longitudinal approaches aim to address

the critical issue of temporal ordering, enhancing the plausibility of causal interpretations regarding mediation effects (Cole & Maxwell, 2003; Georgeson et al., 2023; Gollob & Reichardt, 1991; Gollob & Reichardt, 1987; Maxwell & Cole, 2007; Maxwell et al., 2011; O'Laughlin et al., 2018).

Examining the effects that occur within and between subjects over time presents a significant challenge (Collins & Horn, 1991; Collins & Sayer, 2002). Moreover, the real-world context is where events truly unfold—specifically, it is daily life that reveals the dynamic interplay between variables. Researchers have proposed innovative approaches to address this complexity, including daily diaries, experience sampling methods, and ecological momentary assessment. These research designs involve intensive repeated data collection, enabling a more nuanced understanding of how effects manifest in the everyday lives of individuals (Bolger et al., 2003).

Mediation within the discrete-time vector autoregressive model (DT-VAR), applied to intensive longitudinal data, extends the principles of the cross-lagged panel design. In this approach, variables are regressed on their previous observations (autoregressions) as well as on other variables (crossregressions). However, like cross-lagged panel models, a critical assumption is that the repeated observations of these variables occur at equally spaced intervals (de Haan-Rietdijk et al., 2017). Importantly, the DT-VAR model faces a challenge analogous to that of the cross-lagged panel design: the discrete time-interval used for repeated measurements can impact the estimated coefficients. Consequently, both direct and indirect effects vary based on the specific time-interval chosen for data collection (Gollob & Reichardt, 1991; Gollob & Reichardt, 1987; Reichardt, 2011).

The challenges associated with discrete-time methods can be addressed by adopting a continuous-time modeling framework. Unlike discrete-time approaches, which assume equally spaced data points, continuous-time models accommodate unequally spaced data–a common occurrence in intensive longitudinal studies. These models allow us to capture the evolving dynamics of parameters across a range of time intervals. However, embracing a continuous time perspective necessitates a conceptual shift. Specifically, lagged regression parameters for a given time-interval within this framework are interpreted as total effects, rather than direct effects.

The work of Deboeck and Preacher (2015) provided a starting point for structuring mediation processes in the continuous-time framework. However, their work is far from complete: complete inferences require the availability of reasonable standard error estimates and ways of quantifying the uncertainty around the total, direct, and indirect effects in mediation in continuous time. In this article, we provide estimation procedures to complete the inferential process using the delta and Monte Carlo methods. Additionally, we provide

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the `R` package `cTMed` that calculates point estimates, standard errors, and confidence intervals for the total, direct, and indirect effects. The package also provides plotting tools to visualize how the effects change given a range of time-intervals.

In the following sections, we introduce the linear stochastic differential equation (SDE) framework as a versatile approach for modeling continuous-time processes. Specifically, we delve into a special case of SDE known as the Ornstein-Uhlenbeck (OU) model, which serves as a first-order continuous-time vector autoregressive model (CT-VAR). We present the connection between CT-VAR and DT-VAR, exploring the conceptual shift that occurs when interpreting and calculating total, direct, and indirect effects within the CT-VAR framework. Additionally, we demonstrate how these effects can be computed for arbitrary time intervals. To complete the inferential process, we introduce delta and Monte Carlo methods for estimating standard errors and confidence intervals. To put things in context we provide illustrative examples using simulated and empirical data and provide concrete recommendations moving forward on how to employ the methods we proposed. We end with a discussion of the implications.

Linear Stochastic Differential Equation (SDE) Framework

In our modeling approach, we consider processes that continuously evolve over time. This means that the variables can assume values at any given moment. To capture this behavior, we employ linear stochastic differential equations (SDEs). Specifically, we focus on a special type of SDE known as the Ornstein–Uhlenbeck model (Chow et al., 2023; Oravecz et al., 2011; Uhlenbeck & Ornstein, 1930).

Similar to structural equation modeling, we can break down our model into two key components: the measurement component and the structural component. The measurement model describes the relationship between observed variables and latent (unobserved) variables. Specifically, consider the following equation:

$$\mathbf{y}_{i,t_l} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_{i,t_l} + \boldsymbol{\varepsilon}_{i,t_l}, \quad \text{with} \quad \boldsymbol{\varepsilon}_{i,t_l} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Theta}\right) \tag{1}$$

where $\mathbf{y}_{i,t_l}$, $\boldsymbol{\eta}_{i,t_l}$, and $\boldsymbol{\varepsilon}_{i,t_l}$ are random variables and $\boldsymbol{\nu}$, $\boldsymbol{\Lambda}$, and $\boldsymbol{\Theta}$ are model parameters. $\mathbf{y}_{i,t_l}$ represents a vector of observed random variables, $\boldsymbol{\eta}_{i,t_l}$ a vector of latent random variables, and $\boldsymbol{\varepsilon}_{i,t_l}$ a vector of normally distributed random measurement errors, for time t at measurement occasion l and individual i. $\boldsymbol{\nu}$ denotes a vector of intercepts, $\boldsymbol{\Lambda}$ a matrix of factor loadings, and $\boldsymbol{\Theta}$ the covariance matrix of $\boldsymbol{\varepsilon}$. In the context of a mediation model with three variables where the effect the variable X on the variable Y is mediated by the variable M, we define $\mathbf{y}$ and $\boldsymbol{\eta}$ as follows:

$$\mathbf{y} = \begin{pmatrix} X \\ M \\ Y \end{pmatrix} \quad \text{and} \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_X \\ \eta_M \\ \eta_Y \end{pmatrix}. \tag{2}$$

The dynamic structure is given by

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = -\boldsymbol{\Phi}\left(\boldsymbol{\mu} - \boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t} \tag{3}$$

where $\boldsymbol{\mu}$ is the equilibrium or mean value, $\boldsymbol{\Phi}$ is the drift matrix, $\boldsymbol{\Sigma}$ is the matrix of volatility or randomness in the process, and $\mathrm{d}\boldsymbol{W}$ is a Wiener process or Brownian motion, which

represents random fluctuations. The Ornstein–Uhlenbeck process centers around a central tendency denoted by $\boldsymbol{\mu}$. When the process deviates from this mean, it naturally gravitates back toward it. Think of $\boldsymbol{\mu}$ as an "attractor" that pulls the process back to its typical value. The diagonal elements of the matrix $\boldsymbol{\Phi}$ (referred to as auto effects) control the speed at which the process reverts to its own equilibrium value, $\boldsymbol{\mu}$. When the absolute value of these elements is high, the variable quickly returns to its mean after being perturbed. Smaller absolute values result in slower reversion, allowing the variable to exhibit more persistent behavior away from its mean. The off-diagonal elements of $\boldsymbol{\Phi}$ represent associations between variables and their rates of change. A negative cross effect indicates that if one variable takes on a positive value, there is a resulting decrease in another variable's value. The magnitude of this effect is determined by the absolute value. The diagonal elements of the matrix $\boldsymbol{\Sigma}$ determine the magnitude of volatility around the central tendency $\boldsymbol{\mu}$. Larger values indicate greater fluctuations or noise in the process, while smaller values result in smoother trajectories. Essentially, $\boldsymbol{\Sigma}$ quantifies the degree of randomness or shock-induced variability. The off-diagonal elements of $\boldsymbol{\Sigma}$ represent correlations between random shocks across variables in the system. In summary, the Ornstein–Uhlenbeck process is a mean-reverting stochastic process. Over time, it drifts toward its equilibrium value $\boldsymbol{\mu}$, experiences volatility (as captured by $\boldsymbol{\Sigma}$), and reverts back to the mean with a rate determined by $\boldsymbol{\Phi}$. For the context of this manuscript, we refer to the Ornstein–Uhlenbeck model as the first-order continuous-time vector autoregressive model (CT-VAR).

The drift matrix $\boldsymbol{\Phi}$ is of particular interest to us in the context of mediation models. It encapsulates the associations between the states of the variables we are investigating. Notably, within a trivariate model, a number of possible paths from one variable to another can be explored. This comprehensive connectivity arises because we can specify the model such that all variables are interconnected. Consequently, beyond the classic mediation model $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$, we can also examine alternative mediation pathways, such as $\eta_Y \rightarrow \eta_M \rightarrow \eta_X$, using the same dataset. However, for our current focus, we will delve into the specific $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$ path. In the subsequent section, we will explore the relationship between CT-VAR and DT-VAR models, emphasizing the pivotal role played by the drift matrix $\boldsymbol{\Phi}$.

Link Between CT-VAR and DT-VAR

To understand the connection between CT-VAR and DT-VAR models, we need to examine the discrete-time state space model parameterization of the CT-VAR model which is the discrete-time solution in integral form given by

$$\boldsymbol{\eta}_{i,t_{l_i}} = \boldsymbol{\alpha}_{\Delta t_{l_i}} + \boldsymbol{\beta}_{\Delta t_{l_i}} \boldsymbol{\eta}_{i,t_{l_i-1}} + \boldsymbol{\zeta}_{i,t_{l_i}}, \quad \text{with} \quad \boldsymbol{\zeta}_{i,t_{l_i}} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Psi}_{\Delta t_{l_i}}\right). \tag{4}$$

We first clarify the meaning of the time-interval in both CT-VAR and DT-VAR models. In the context of CT-VAR models, the underlying process can take values at any arbitrary time point. However, our observations occur discretely. This means that while the process itself operates continuously, we only observe it at specific discrete time instances. Let's compare this to DT-VAR models, where observations are made at specific discrete time intervals (usually positive integers). In a DT-VAR example, suppose we collect data every hour from 7 AM to 10 AM for four measurement occasions. We make discrete observations in time

t, scaled by the hour, on specific measurement occasions $l = \{1 \cdots 4\}$. These occasions correspond to time points: $t_1 = 7$, $t_2 = 8$, $t_3 = 9$, and $t_4 = 10$. The time-interval from one measurement occasion to the next is constant and equal to 1 hour. In the CT-VAR model, observations can occur at any time between 7 AM and 10 AM for the same four measurement occasions. Specifically, the time points are given by: $t_1 = 7.166667$ (equivalent to 7:10 AM), $t_2 = 8.283333$ (equivalent to 8:17 AM), $t_3 = 9.45$ (equivalent to 9:27 AM), and $t_4 = 10$ (equivalent to 10:00 AM). Notably, the time-interval from one measurement occasion to the next varies, allowing for irregularly spaced observations. We denote this time-interval as Δt, representing the difference between the current measurement occasion l and the previous measurement occasion $l-1$. From here we can see that if Δt is constant as in a DT-VAR model, the model is simply a CT-VAR model where Δt is constant. As such, the DT-VAR model is a special case of the CT-VAR model.

Going back to Equation 4, we see that $\boldsymbol{\eta}_{i,t_{l_i}}$, $\boldsymbol{\eta}_{i,t_{l_i-1}}$, and $\boldsymbol{\zeta}_{i,t_{l_i}}$ are random variables while $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\Psi}$ are model parameters. The model parameter $\boldsymbol{\alpha}$ denotes a vector of intercepts, $\boldsymbol{\beta}$ a matrix of coefficients describing the impact of the lagged values $\boldsymbol{\eta}_{i,t_{l-1}}$ on the current value $\boldsymbol{\eta}_{i,t_l}$ (autoregression coefficients on the diagonal and cross regression coefficients on the off-diagonal), and $\boldsymbol{\Psi}$ the covariance matrix of $\boldsymbol{\zeta}$.

Let us look at the link between the CT-VAR and the discrete-time solution parameters. Whereas, the parameter $\boldsymbol{\Phi}$ describes the moment-to-moment association of one variable on the rate of change in another, the parameter $\boldsymbol{\beta}$ is a matrix of lagged parameters, that is, the effect of the variable on a previous measurement occasion to the current measurement occasion, given a particular time-interval Δt. SHOULD I WRITE MORE ABOUT THE MEANING OF $\boldsymbol{\beta}$? WHAT OTHER ASPECTS OF $\boldsymbol{\beta}$ AND THE ASSOCIATION WITH $\boldsymbol{\Phi}$ SHOULD BE INCLUDED. The discrete-time $\boldsymbol{\beta}$ can be obtained from the continuous-time $\boldsymbol{\Phi}$ using

$$\boldsymbol{\beta} = \exp\left(\Delta t \boldsymbol{\Phi}\right). \tag{5}$$

Furthermore, the discrete-time parameter $\boldsymbol{\alpha}$ is related to the continuous-time parameters $\boldsymbol{\Phi}$ and $\boldsymbol{\mu}$

$$\begin{aligned} \boldsymbol{\alpha} &= \left(\boldsymbol{\Phi}^{-1}\right)\left(\exp\left(\Delta t \boldsymbol{\Phi}\right) - \mathbf{I}\right)\left(-\boldsymbol{\Phi}\boldsymbol{\mu}\right) \\ &= \left(\boldsymbol{\Phi}^{-1}\right)\left(\boldsymbol{\beta} - \mathbf{I}\right)\left(-\boldsymbol{\Phi}\boldsymbol{\mu}\right). \end{aligned} \tag{6}$$

If we solve for $\boldsymbol{\mu}$ in Equation 6, we arrive at

$$\boldsymbol{\mu} = \left(\boldsymbol{I} - \boldsymbol{\beta}\right)^{-1}\boldsymbol{\alpha}. \tag{7}$$

Assuming weak stationarity, the mean vector of $\boldsymbol{\eta}$ as a function of the discrete-time parameters $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ is given by $\boldsymbol{\mu}_{\boldsymbol{\eta}} = \left(\mathbf{I} - \boldsymbol{\beta}\right)^{-1}\boldsymbol{\alpha}$. This is equivalent to the CT-VAR parameter $\boldsymbol{\mu}$ as a function of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ given by Equation 7.

Lastly, we can convert the continuous-time process noise covariance matrix $\boldsymbol{\Sigma}$ to the discrete-time counter part $\boldsymbol{\Psi}$[1] WHAT IS THE INTERPRETATION OF THE PROCESS NOISE IN DT-VAR COMPARED TO CT-VAR? ARE THEY THE SAME?

[1]Note that there are many equivalent notations used in the literature. This notation lends itself to calculation in statistical packages that allow for standard linear algebra operations.

$$\text{vec}\left(\boldsymbol{\Psi}\right) = \left[\left(\boldsymbol{\Phi} \otimes \mathbf{I}_p\right) + \left(\mathbf{I}_p \otimes \boldsymbol{\Phi}\right)\right]\left[\exp\left(\left[\left(\boldsymbol{\Phi} \otimes \mathbf{I}_p\right) + \left(\mathbf{I}_p \otimes \boldsymbol{\Phi}\right)\right]\Delta t\right) - \mathbf{I}_{p\times p}\right]\text{vec}\left(\boldsymbol{\Sigma}\right). \quad (8)$$

In summary, we can see the DT-VAR parameters are related to CT-VAR parameters. We can also see from the discrete state space integral solution of CT-VAR that DT-VAR is a special case of CT-VAR where the time-interval Δt is constant across time and individuals. In the next section, we will delv deeper on the matrix of lagged parameters $\boldsymbol{\beta}$ as this parameter matrix is what is of interest in mediation models. We will look at how to define the total, direct, and indirect effects in the current modeling framework.

Rethinking the Interpretation of the β Matrix as the Total Effects Matrix

In a typical recursive path model, the simple mediation model, where the independent variable X and the dependent variable Y are related through a mediator variable M, is given by the following linear equations

$$Y = \delta_1 + cX + \varepsilon_1, \quad (9)$$

$$M = \delta_2 + aX + \varepsilon_2, \quad \text{and} \quad (10)$$

$$Y = \delta_3 + c'X + bM + \varepsilon_3. \quad (11)$$

The total effect of X on Y is represented by c, the direct effect of X on Y controlling for M is represented by c', and the indirect effect which is the effect of X on Y through M is the difference between the total and direct effects $(c - c')$. Equivalently, using path-tracing rules, the indirect effect is the product of a and b $(ab = c - c')$, that is, the product of the paths we used to traverse X to Y through M. In causal terms, if X changes by one unit, then M changes by a units as a result similarly if M changes by one unit, then Ychanges by b units as a result. The resulting change of a units on M produced by a unit change in X would then produce a corresponding change of b units on Y. Thus Y changes by ab units through M as X changes by one unit.

In both recursive path models and autoregressive models, the fundamental concept involves decomposing the total effect into direct and indirect components. This approach allows us to understand how the mediator variable M influences the relationship between X and Y. However, in the context of autoregressive models, several authors (e.g., Aalen et al., 2016; Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) propose a conceptual shift when thinking about the lagged parameters $\boldsymbol{\beta}$. In this conceptualization, $\boldsymbol{\beta}$ is treated as the matrix of total effects rather than direct effects. This shift has implications for thinking about mediation models in DT-VAR particularly when it comes to decomposing the total effect into direct and indirect effects.

To illustrate this point, assume that data comes from a CT-VAR process described previously. We use the observed and latent variable vectors for three observed variables X, M, and Y in Equation 2. Let t be time in hours and $\Delta t = 1$ a time-interval of one hour between measurement occasions given by Figure 1a. Given Equation 5 we can calculate the matrix of lagged coefficients for a time-interval $\Delta t = 1$ from the drift matrix $\boldsymbol{\Phi}$ as follows:

$$\boldsymbol{\beta}_{\Delta t(1)} = \exp\left(\boldsymbol{\Phi}\right). \tag{12}$$

Let the elements of $\boldsymbol{\beta}_{\Delta t(1)}$ be given by

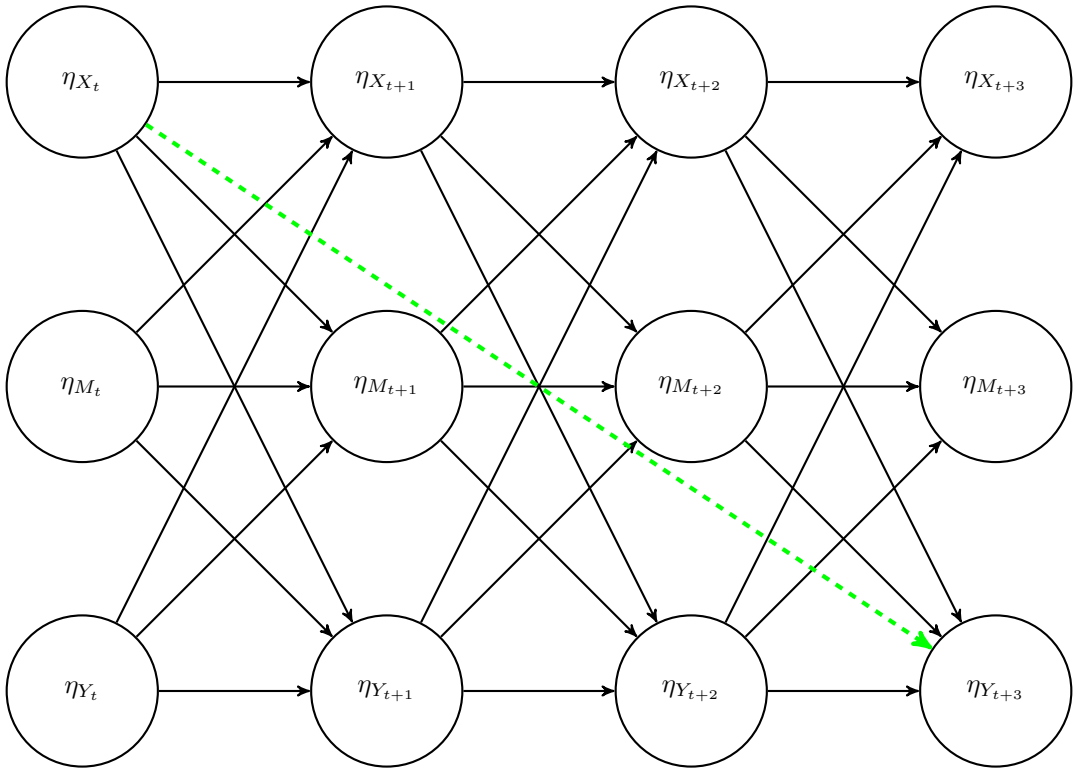
$$\boldsymbol{\beta}_{\Delta t(1)} = \begin{array}{c} \\ \eta_{X_{t+1}} \\ \eta_{M_{t+1}} \\ \eta_{Y_{t+1}} \end{array} \begin{array}{c} \begin{array}{ccc} \eta_{X_t} & \eta_{M_t} & \eta_{X_t} \end{array} \\ \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \beta_{1,3} \\ \beta_{2,1} & \beta_{2,2} & \beta_{2,3} \\ \beta_{3,1} & \beta_{3,2} & \beta_{3,3} \end{pmatrix} \end{array}. \tag{13}$$

Now we look at the case of a DT-VAR model from the same continuous process except that the time-interval used is three hours ($\Delta t = 3$) between measurement occasions given by Figure 1b. From the time-series literature (e.g., Hamilton, 1994) we can derive $\boldsymbol{\beta}_{\Delta t(3)}$ from $\boldsymbol{\beta}_{\Delta t(1)}$ using matrix powers. The key is to raise the matrix with the smaller time-interval by the factor in which the time-interval increased to the larger time-interval. For example, the change from $\Delta t = 1$ to $\Delta t = 3$ is a factor of three, thus $\boldsymbol{\beta}^3_{\Delta t(1)} = \boldsymbol{\beta}_{\Delta t(3)}$. Thus, we can say that $\boldsymbol{\beta}_{\Delta t(3)} = \exp\left(3\boldsymbol{\Phi}\right) = \boldsymbol{\beta}^3_{\Delta t(1)}$. Let us explore the relationship of the elements of $\boldsymbol{\beta}_{\Delta t(3)}$ as a function of the elements of $\boldsymbol{\beta}_{\Delta t(1)}$. In particular, we will focus on the third row and first column in $\boldsymbol{\beta}_{\Delta t(3)}$ pertaining to the association of η_X at time t and η_Y at time $t+3$. The green line in Figure 1b represents the total effect of η_X at time t and η_Y at time $t+3$. This total effect is decomposed as a function of the lagged parameters in $\boldsymbol{\beta}_{\Delta t(1)}$ presented in Figure 2. By raising Equation 13 to the third power and subsetting the third row and the first column of the resulting matrix we get

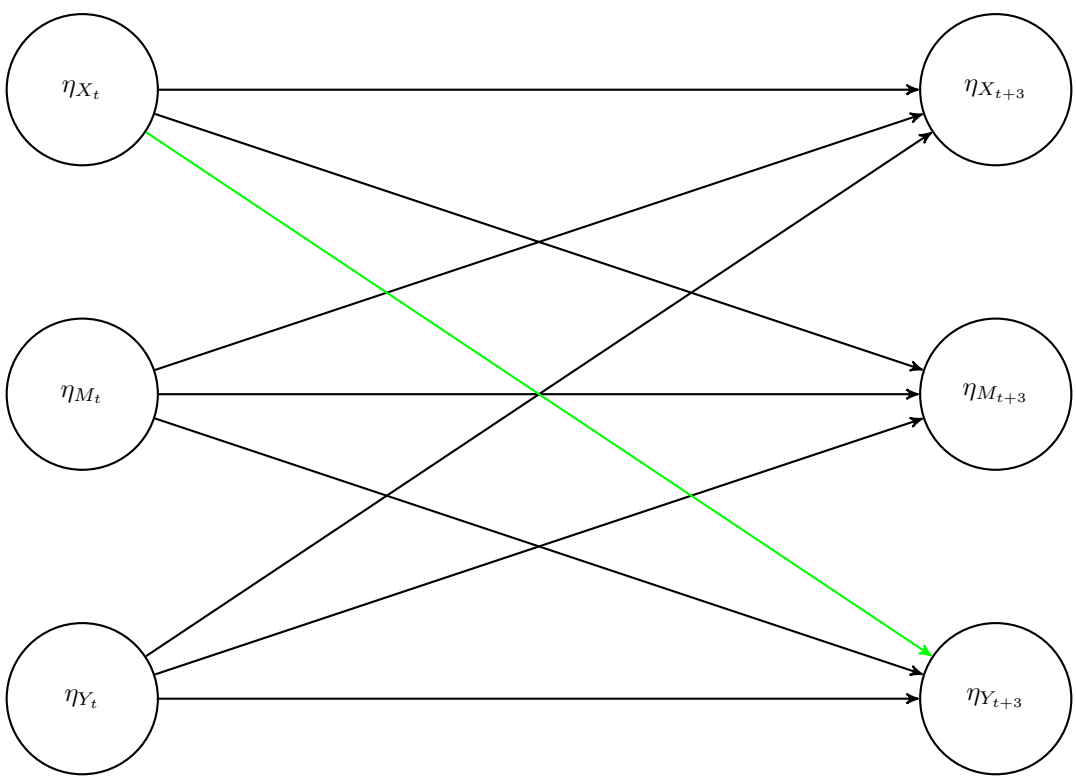
$$\begin{aligned} \boldsymbol{\beta}_{\Delta t(3)}\left[3,1\right] = {} & \beta_{1,1}\beta_{1,1}\beta_{3,1} + \beta_{2,1}\beta_{1,2}\beta_{3,1} + \beta_{3,1}\beta_{1,3}\beta_{3,1} \\ & + \beta_{1,1}\beta_{2,1}\beta_{3,2} + \beta_{2,1}\beta_{2,2}\beta_{3,2} + \beta_{3,1}\beta_{2,3}\beta_{3,2} \\ & + \beta_{1,1}\beta_{3,1}\beta_{3,3} + \beta_{2,1}\beta_{3,2}\beta_{3,3} + \beta_{3,1}\beta_{3,3}\beta_{3,3}, \end{aligned} \tag{14}$$

where each term represents distinct paths from η_X at time t to η_Y $t+3$ graphically represented by each panel in Figure 2. Note that red paths in are paths where η_M is traversed. We define these red paths as indirect effects of η_X at time t on η_Y at time $t+3$ through η_M. The sum of these red paths constitute the overall indirect effect. The blue paths are the direct effects and their sum constitutes the overall direct effect of η_X at time t on η_Y at time $t+3$.

In summary, we can say that the matrix of lagged coefficients $\boldsymbol{\beta}$ for a particular time-interval Δt is the sum of paths at smaller time-intervals. As such, it is conceptually the total effect. Related to this is the issue of time-interval dependency, that is, given the same underlying continuous process that generates data for the variables, the matrix of lagged coefficients will change depending on the time-interval at which the variables are observed. As the direct and indirect effect are simply a decomposition of the total effect, the substantive interpretation of the mediation model changes as a function of the time-interval. The time-interval is so important in DT-VAR such that it is a requirement that variables are measured at regular time-intervals. However, intensive longitudinal data can often be collected in irregularly spaced measurements by design. For example, partially random collection of data is often employed using beepers and mobile devices. Data collection



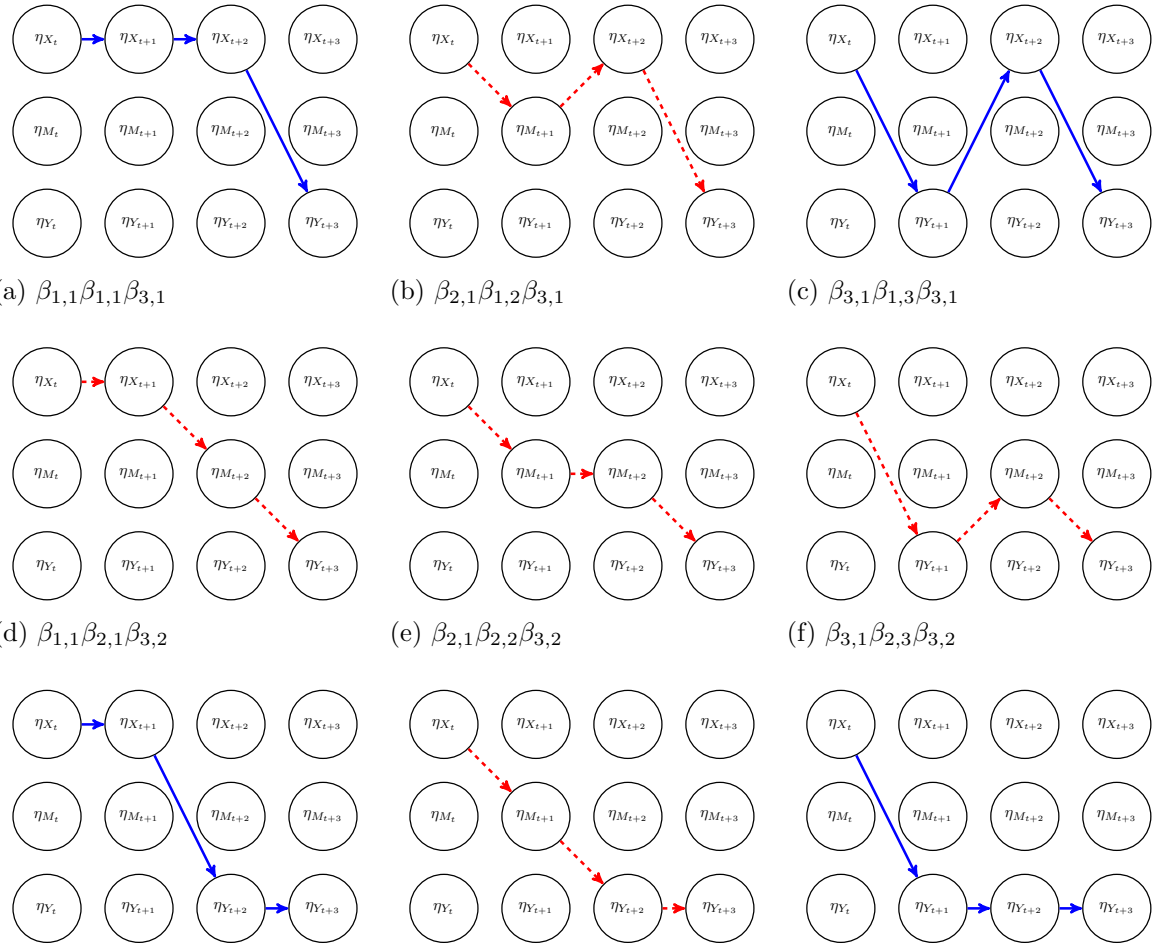
(a) *DT-VAR with time-interval of one for four measurement ocassions.*



(b) *DT-VAR with time-interval of three. The green line represents the total effect of X on Y at a time-interval of three.*

Figure 1

Lagged Parameters in DT-VAR for (a) time-interval of 1 and (b) time-interval of 3.



(a) $\beta_{1,1}\beta_{1,1}\beta_{3,1}$ (b) $\beta_{2,1}\beta_{1,2}\beta_{3,1}$ (c) $\beta_{3,1}\beta_{1,3}\beta_{3,1}$

(d) $\beta_{1,1}\beta_{2,1}\beta_{3,2}$ (e) $\beta_{2,1}\beta_{2,2}\beta_{3,2}$ (f) $\beta_{3,1}\beta_{2,3}\beta_{3,2}$

(g) $\beta_{1,1}\beta_{3,1}\beta_{3,3}$ (h) $\beta_{2,1}\beta_{3,2}\beta_{3,3}$ (i) $\beta_{3,1}\beta_{3,3}\beta_{3,3}$

Figure 2

The path from η_X to η_Y for DT-VAR with time-interval of three as a function of paths from η_X to η_Y in DT-VAR with time-interval of one. Red paths indicate indirect effects. Blue paths indicate direct effects.

can also be contingent on specific behaviors that individuals engage in. These approaches offer a glimpse into the real-world experience, enhancing ecological validity. However, they result in irregularly spaced data. The discrepancy between the data collection scheme and methodological requirements of DT-VAR specifically regular spacing can result in erroneous conclusions regarding the nature of the system being studied. This underscores the value of the CT-VAR as a way of getting at the total, direct, and indirect effects in a mediation model.

In the next section, we will discuss how to derive the total, direct, and indirect effects from the drift matrix $\boldsymbol{\Phi}$ for any arbitrary time-interval Δt. We will discuss two approaches, namely, the unidirectional and bidirectional approaches. We propose the bidirectional approach, present how to calculate the total, direct, and indirect effects given a drift matrix $\boldsymbol{\Phi}$ in R as well as plotting options.

Deriving the Total, Direct, and Indirect Effects from the Drift Matrix $\boldsymbol{\Phi}$ for Time-Interval Δt

Now we divide any arbitrary time-interval Δt into infinitely many smaller parts s. Let s be the limit as $l \to \infty$. This can be visualized in Figure 3. Similar to our previous illustration, we can say that the matrix of lagged parameters for any arbitrary time-interval Δt is the total effects matrix represented by Figure 3.a. The total effects can be decomposed into the direct and indirect effects presented in Figure 3.b. For example, the overall direct effect of the variable X at time t on the variable Y at time $t + \Delta t$ is equal to all the paths from X at time t to Y at time $t + \Delta t$ that does not pass through M.

Unidirectional Approach

DESCRIBE HERE THE APPROACH BY DEBOECK AND PREACHER AND SHOW THE LIMITATIONS.

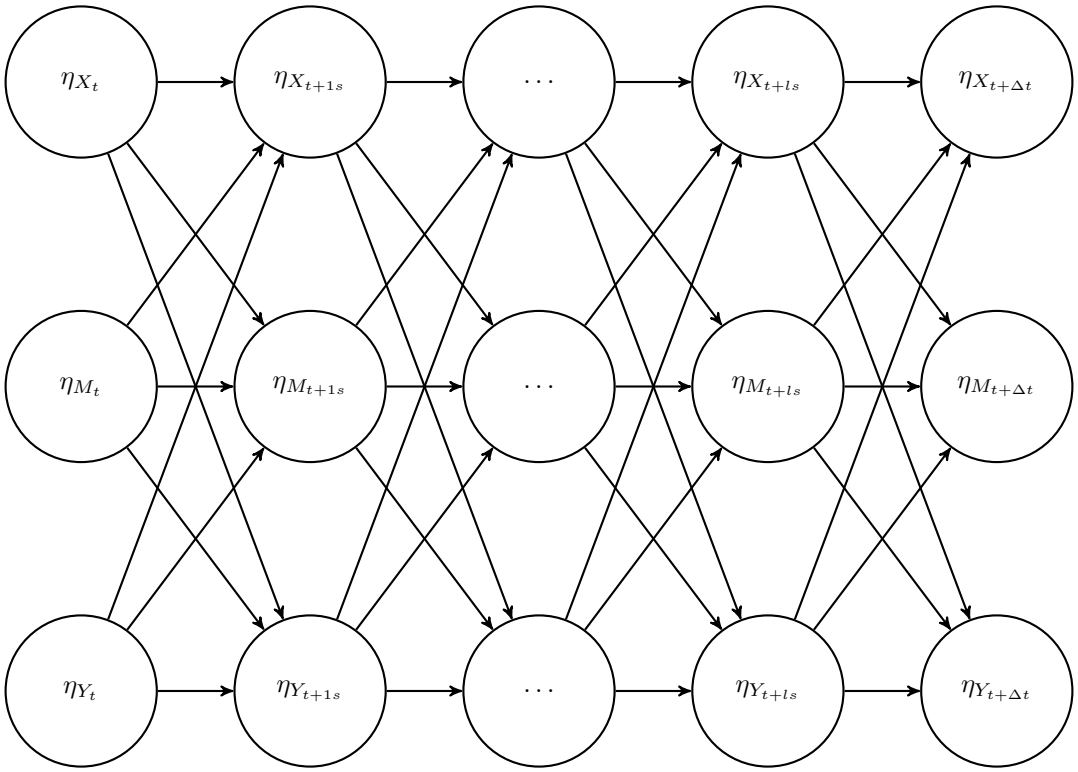
Bidirectional Approach

PUT THE EQUATIONS IN WORDS.

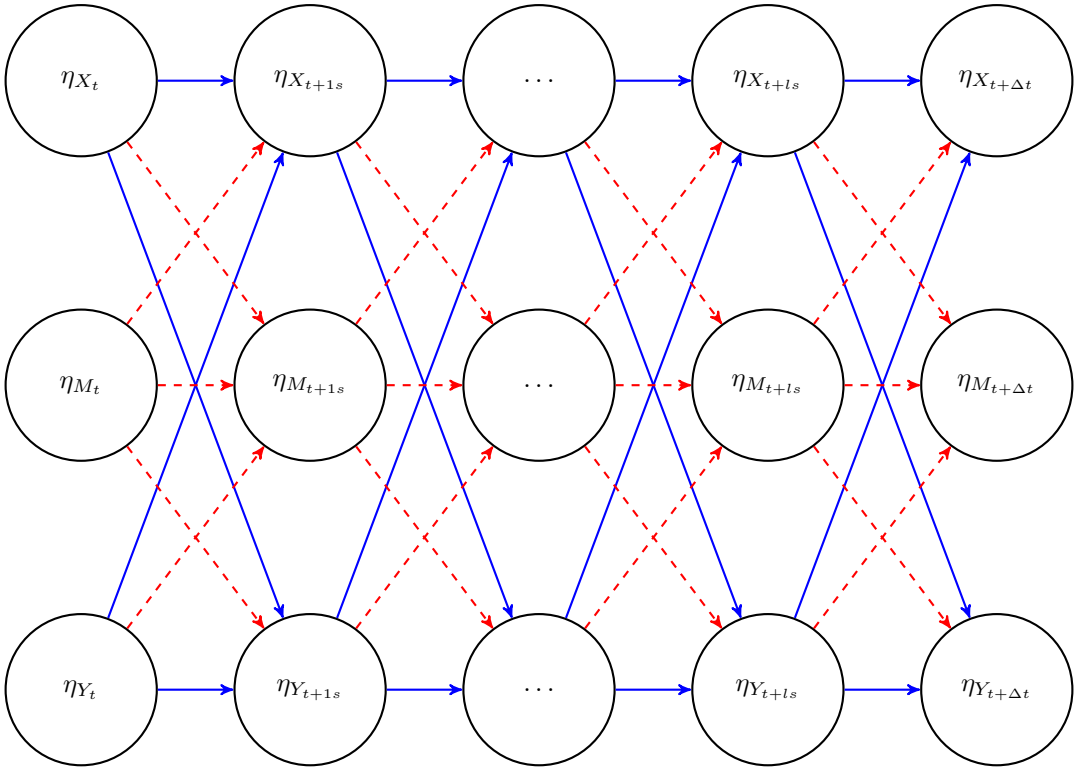
$$\text{Total} = \exp\left(\Delta t \boldsymbol{\Phi}\right) \tag{15}$$

$$\text{Direct} = \exp\left(\Delta t \boldsymbol{D_m} \boldsymbol{\Phi} \boldsymbol{D_m}\right) \tag{16}$$

$$\text{Indirect} = \exp\left(\Delta t \boldsymbol{\Phi}\right) - \exp\left(\Delta t \boldsymbol{D_m} \boldsymbol{\Phi} \boldsymbol{D_m}\right) = \text{Total} - \text{Direct} \tag{17}$$



(a) *Total Effects*



(b) *Direct (Solid Blue Lines) and Indirect (Dashed Red Lines) Effects*

Figure 3

Effects Decomposition from a CT-VAR Perspective

MAYBE PUT THIS SOMEWHERE ELSE? PUT IT WHEN DISCUSSING UNCERTAINTY AROUND ESTIMATES?
Estimating the indirect effect in a mediation model, although conceptually straightforward as a product of regression coefficients, presents unique challenges. The sampling distribution of the mediation effect becomes asymmetric if it deviates from exactly zero in the population (Aroian, 1947; Craig, 1936), leading to complexities in computing p-values for the indirect effect. Consequently, researchers commonly rely on confidence intervals (CIs) to make inferences about the indirect effect. Early methods, such as the normal-theory approach, assume symmetric sampling distributions (Aroian, 1947; Goodman, 1960; Sobel, 1982, 1986, 1987). However, alternative approaches address the asymmetry, including: (1) The distribution of the product method, which analytically examines the sampling distribution of the product of normally distributed coefficients (MacKinnon et al., 2004, 2007); (2) Nonparametric bootstrapping, involving resampling the original data and fitting the model multiple times to create a sampling distribution of the indirect effect (Bollen & Stine, 1990; Preacher & Hayes, 2008; Shrout & Bolger, 2002); (3) Parametric bootstrapping, involving simulating raw data using the estimated parameters and fitting the model similar to nonparametric bootstrapping (Preacher & Selig, 2012); (4) The Monte Carlo method, which simulates regression coefficients to generate a sampling distribution of the indirect effect (MacKinnon et al., 2004; Pesigan & Cheung, 2023; Preacher & Selig, 2012); (5) Profile likelihood-based approach that estimate CIs using the likelihood ratio test (Cheung, 2009a, 2009b; Pawitan, 2013; Pesigan & Cheung, 2020; Venzon & Moolgavkar, 1988). Nonparametric bootstrapping has emerged as the preferred method for generating confidence intervals (CIs) when estimating the indirect effect in mediation models [ADD CITATIONS HERE]. However, the multiple model fitting especially for more complex models can be a bottleneck making it a less viable approach (see Pesigan & Cheung, 2023, for the case of mediation models with missing data). The Monte Carlo method has been shown to be as good as nonparametric bootstrap in a number of situations (Pesigan & Cheung, 2023; Tofighi & Kelley, 2019, 2020). Because the empirical sampling distributions of parameter estimates are generated and not obtained via multiple model fitting, the Monte Carlo method is computationally more efficient than nonparametric bootstrapping.

Standard Errors and Confidence-Intervals for Total, Direct, and Indirect Effects

Delta Method Confidence Intervals

Let $\boldsymbol{\theta}$ be $\text{vec}\left(\boldsymbol{\Phi}\right)$, that is, the elements of the $\boldsymbol{\Phi}$ matrix in vector form sorted columnwise. Let $\hat{\boldsymbol{\theta}}$ be $\text{vec}\left(\hat{\boldsymbol{\Phi}}\right)$. By the multivariate central limit theory, the function $\mathbf{g}$ using $\hat{\boldsymbol{\theta}}$ as input can be expressed as:

$$\sqrt{n}\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right) - \mathbf{g}\left(\boldsymbol{\theta}\right)\right) \xrightarrow{\text{D}} \mathcal{N}\left(0, \mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right) \tag{18}$$

where $\mathbf{J}$ is the matrix of first-order derivatives of the function $\mathbf{g}$ with respect to the elements of $\boldsymbol{\theta}$ and $\boldsymbol{\Gamma}$ is the asymptotic variance-covariance matrix of $\hat{\boldsymbol{\theta}}$. From the former, we can derive the distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ as follows:

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right) \approx \mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right), n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right) \tag{19}$$

The uncertainty associated with the estimator $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is, therefore, given by $n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'$. When $\boldsymbol{\Gamma}$ is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of $\hat{\boldsymbol{\theta}}$, that is, $\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)$ for $n^{-1}\boldsymbol{\Gamma}$. Therefore, the sampling variance-covariance matrix of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is given by

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right) \approx \mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right), \mathbf{J}\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\mathbf{J}'\right). \tag{20}$$

From this general approach, we can generate the sampling variance-covariance matrix of any function of $\boldsymbol{\Phi}$. The `DeltaMed` function in the `cTMed` package applies the delta method to generate standard errors, test statistics, p-values and confidence intervals for the total, direct, and indirect effects in continuous time.

Monte Carlo Method Confidence Intervals

Based on the asymptotic properties of maximum likelihood estimators, we can assume that they are normally distributed around the population parameters.

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}\left(\boldsymbol{\theta}, \mathbb{V}\left(\hat{\boldsymbol{\theta}}\right)\right) \tag{21}$$

Using this distributional assumption, a sampling distribution of $\hat{\boldsymbol{\theta}}$ which we refer to as $\hat{\boldsymbol{\theta}}^*$ can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\boldsymbol{\theta}}^* \sim \mathcal{N}\left(\hat{\boldsymbol{\theta}}, \hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\right). \tag{22}$$

A sampling distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$, which we refer to as $\mathbf{g}\left(\hat{\boldsymbol{\theta}}^*\right)$, can be generated by using the simulated estimates to calculate $\mathbf{g}$. The standard deviations of the simulated estimates are the standard errors. Percentiles corresponding to $100\left(1-\alpha\right)\%$ are the confidence intervals.

From this general approach, we can generate the empirical sampling distribution of any function of $\boldsymbol{\Phi}$. There are constraints, however, on the randomly generate values of $\boldsymbol{\Phi}$. First, the diagonal values should be from zero to negative infinity. Second, the real part of all eigenvalues is less than zero. These two can be implemented using a replacement sampling approach, that is, discarding randomly generated values where the two constraints do not hold. The `MCMed` function in the `cTMed` package applies the Monte Carlo method to generate the standard errors and confidence intervals for the total, direct, and indirect effects in continuous time.

Setting Up the Problem

Quantifying Uncertainty in Direct, Indirect, and Total Effects in Unidirectional Mediation Context

Taking the bidirectional framework presented earlier and implementing it within a "benchmark" example presented by Deboeck and Preacher. Goal: illustrate how uncertainty in direct, indirect, and total effects can be quantified using confidence intervals constructed

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Figure 4

Data Generating Model Used in Deboeck and Preacher (2015)

using the delta and MC methods at $\Delta t = 1$ and 2, the particular time intervals considered by D & P.

Monte Carlo simulation study: compare the performance of the two CI estimation approaches under conditions with: (1) different sample sizes; (2) different Δt values; and (3) different magnitudes of indirect effects. Include table/plot illustrating what the empirical sampling distribution of indirect effect looks like, the average percentile intervals ("true" confidence intervals) in comparison to the CIs based on the delta method and the MC method.

Include table of MC simulation results highlighting which approach did better in terms of coverage, margin of error. Are the conclusions about the significance of the indirect and direct effects different at any Δt under the two approaches.

In this example, we simulated data following the approach outlined by Deboeck and Preacher (2015). Specifically, we generated data from a DT-VAR model using a fixed time interval of one unit. The three variables—X, M, and Y—were measured at seven distinct time points. The data generating model is visually depicted in Figure 1.

However, unlike Deboeck and Preacher's work, which employed an extremely large sample size ($n = 10,000$), we intentionally used a smaller sample size of $n = 50$. Our goal was to explore the finite sample performance of the methods described in the literature. To generate the data, we utilized the `simStateSpace` `R` package. Detailed information about the data generation process can be found at this link (https://jeksterslab.github.io/cTMed/articles/data.html).

Before fitting our model, we oversampled the data to achieve a finer time resolution ($\Delta t = 0.10$) by inserting missing values between the empirical time intervals. The model itself was estimated using the `dynr` `R` package (Ou et al., 2019). To generate confidence intervals around the estimate effects, we employed both the delta and Monte Carlo methods. Specifically, we explored a range of time-interval values from 0 to 20[2]. Our focus was on showing how the total, direct, and indirect effects change as a function of the time interval. Detailed instructions on generating confidence intervals using the delta method and Monte Carlo simulations can be found in the `cTMed` `R` (https://jeksterslab.github.io/cTMed/articles/delta.html and https://jeksterslab.github.io/cTMed/articles/mc.html).

Deboeck and Preacher (2015) previously described the estimates of effects for the model at hand. Our focus now shifts to the confidence intervals presented in Figure X. Notably, these confidence intervals exhibit varying widths across different time intervals. The widest confidence intervals occur at the minimum and maximum points of the curve for all methods. As mentioned in the introduction, the delta method assumes normality

[2]Strictly speaking, when a value of 0 is used as an input in DeltaMed and MCMed, these functions utilize the smallest double precision number available in the system, i.e., `.Machine$double.xmin`.

in the sampling distribution. Consequently, it imposes symmetry on confidence intervals. This symmetry is evident in the plots. In contrast, the Monte Carlo method allows for asymmetry. Specifically, this asymmetry is visible in the tail-ends of the curves.

The distinction between thw delta and Monte Carlo methods is crucial because functions of normally distributed parameters can yield non-normal sampling distributions. To illustrate this further, we conducted simulations using $1,000$ data sets based on the previously described model. For each data set, we estimated the drift matrix. From these matrices, we calculated the total, direct, and indirect effects for a time interval of 15. Figure X reveals that while the total effect exhibits relative symmetry, the direct and indirect effects deviate from normality, particularly in the tails. Specifically, the indirect effect skews to the right and the direct effect skews to the left. These findings highlight the importance of considering both symmetry and asymmetry in confidence intervals when analyzing DT-VAR models.

We have established that both the delta method and the Monte Carlo method are viable approaches for generating confidence intervals (CIs). These methods play a crucial role in quantifying uncertainty around parameter estimates. Our investigation revealed an important distinction: the delta method imposes symmetry on confidence intervals, while the Monte Carlo method allows for asymmetry. This flexibility is particularly relevant when dealing with non-normal sampling distributions. We demonstrated that the sampling distribution of functions derived from the drift matrix can deviate from normality. This finding underscores the need to consider alternative methods that account for such deviations. In the next section, we will delve deeper by comparing the performance of both methods–specifically, examining confidence interval coverage in a small-scale simulation study.

Example 2: Comparison of Uncertainty Quantification of Effects Using the Delta and Monte Carlo Methods

Deboeck and Preacher (2015) introduced continuous-time vector autoregressive (CT-VAR) models as a powerful tool for analyzing traditional cross-lagged panel data. However, the benefits of CT-VAR extend beyond this context. In particular, CT-VAR proves advantageous for intensive repeated samples, such as those encountered in ecological momentary assessment (EMA) studies, especially when dealing with irregularly spaced observations. To demonstrate the versatility of CT-VAR, we present a simulation study that emulates an EMA scenario. In EMA studies, researchers often collect data at irregular intervals over short time spans. To showcase the applicability of CT-VAR, we generated synthetic data resembling an EMA study. We simulated observations for three variables–X, M, and Y. The time vector spanned from 0 to 24×7 (representing hourly measurements within a day over seven days). Observations were randomly selected, resulting in 100 data points. Unselected observations were flagged as missing values, creating irregularly spaced data[3].

As for the parameters, we employed the drift matrix used by Deboeck and Preacher (2015). Additionally, we transformed the DT-VAR process noise covariance matrix from Example 1 into its CT-VAR counterpart. Our approach involved converting the DT-VAR parameters to SVAR form. By leveraging the SVAR contemporaneous effects and the process

[3]Note that the initial value is always included in the sample such that all individuals have the same starting point in time.

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Figure 5

Simulation Results Comparing the Coverage Probabilities of the Delta Method and the Monte Carlo Method Confidence Intervals

noise covariance matrix, we derived the CT-VAR process noise covariance matrix, following the methodology outlined by Chow et al. (2023). To generate the data, we utilized the `simStateSpace` `R` package. Detailed information about the data generation process can be found at this link (https://jeksterslab.github.io/cTMed/articles/ADD-LINK-HERE.html). Subsequently, we employed the `dynr` `R` package to fit the CT-VAR model, similar to Example 1. Notably, in the current case, we incorporated a measurement model component. For further details, please refer to this resource (https://jeksterslab.github.io/cTMed/articles/ADD-LINK-HERE.html).

We assessed the performance of the delta method and the Monte Carlo method for constructing confidence intervals. Our focus was on the coverage probability, which represents the proportion of simulation replications where the confidence intervals contained the true population parameter. We employed a 95% confidence interval as our benchmark. A method is considered robust if its coverage probability falls within the range of 0.925 to 0.975, as suggested by (Bradley, 1978). Results are pressented in Figure 5. For both total and direct effects, both the delta method and the Monte Carlo method met the robustness criteria. However, when examining the indirect effect, the delta method exhibited a drop in coverage probability to 0.89 at a time interval of 20. Notably, a noticeable decline began at a time interval of 15 for the delta method. The observed behavior can be attributed to the delta method's tendency to impose symmetry on confidence intervals, even when the underlying sampling distribution is non-symmetric. I HAVE THE DATA TO SHOW THIS BUT I AM NOT SURE HOW. CAN I SHOW THAT THE SKEWNESS MEASURE OF THE SAMPLING DISTRIBUTION IS CORRELATED WITH COVERAGE PROBABILITY FOR THE DELTA METHOD? WHAT OTHER WAYS OF LOOKING AT THE SIMULATION RESULTS CAN SHOW EMPIRICALLY THE ISSUES ARRISING FROM THE DELTA METHOD THAT MC CAN ADDRESS?

I HAVE RESULTS FOR DIFFERENT SAMPLE SIZES 50, 100, 150, AND 200 BUT I HAVE ONLY DOWNLOADED AND SUMMARIZED N 50 FROM ROAR COLLAB.

Example 3: Empirical Illustration and Conclusions

In this section we illustrate the application of the delta and Monte Carlo methods on an empirical data set.

Key aim is to address the question of whether the indirect effect of self-doubt on fatigue, as mediated reciprocally by irritation and restlessness, is significantly different from zero. Discrepancies between conclusions based on the delta method and MC method.

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(a) *Example from Deboeck and Preacher (2015)*

.setup/latex/figures/pdf/ct-med-empirical.pdf

(b) *Empirical Example from Ryan and Hamaker (2021)*

Figure 6

Total, Direct, and Indirect Effects as a Function of Time-Interval

Data

I used the $n = 1$ empirical example in Ryan and Hamaker. I used both ctsem and dynr. The estimates are different but kinda close enough.

Open Practices Statement

The data and materials for this study are available on OSF (https://osf.io/zae7s/?view_only=855be2012b554f05bbc679946845bab8) and GitHub (https://github.com/jeksterslab/manCTMed, https://jeksterslab.github.io/manCTMed/index.html).

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